

ClassLedger: Product Requirements

A student and teacher management tool for a small coaching class, with AI support.

Working name: ClassLedger. Clickable prototype:

<https://claude.ai/artifact/WM4afjkfQ66BR1rgD7ftyb>

1. What we are building

One line: a simple tool for a coaching class that tracks attendance, marks, fees and topic-wise performance, and tells the head teacher which students need attention.

Why it is different from an LMS: an LMS delivers content (courses, assignments). We manage the academic record and act on it: red flags, chapter and topic analysis, and AI-made question papers. The first version does not replace any LMS.

Target customer (assumed): a coaching class with one head teacher who is also the owner and teaches Maths, plus 2 subject teachers (Chemistry, Physics). Students attend all three subjects. There may be several batches.

2. Roles (version 1)

Role	Who	Access
Head teacher	Owner, teaches Maths	Everything, all batches and subjects
Student	Enrolled student	Own record only

Not in version 1: a separate Chemistry and Physics teacher login. Until then the head teacher does the data entry. Add a Teacher role later (own subject fully, other subjects read-only).

Privacy rule: a student sees the batch average but never other students' names or marks. Enforce this in the backend, not only on screen.

3. Head teacher screens

Four tabs: **Overview** (default), **Students**, **Tests**, **Fees**.

3.1 Overview

- Today's attendance:** present %, on time, late, absent. Sessions for Maths, Chemistry and Physics, each marked or not marked, with a "Mark now" button.
- Batch-wise analysis:** per batch: students, present today, monthly attendance, average marks in each subject, weakest subject.
- Red flag students:** name (linked to profile), batch, reason, average, call-parent button.
- Notice board:** tests, holidays, off days. Head teacher posts notices; students see them read-only.
- Also: pending work (attendance not marked, marks missing) and quick actions (add student, create test, enter marks, export report).

3.2 Students tab

- Two searches: **by batch name** and **by student name** (they work together).

- Quick-view columns, no marks: **Today** (present, late, absent), **Attendance**, **Trend** (improving, steady, dropping), **Fees**, **Flag**.
- Every name links to the student profile.
- "Add student" button.

3.3 Student profile (teacher view)

- Name, batch, parent phone, Call parent, Send progress report.
- Tiles: overall attendance, average, today's status, fee status.
- Red flag reasons.
- Subject cards: latest score, last 4 scores, change since last test, difference from batch average, subject attendance.
- **Chapter and topic analysis** (see section 5).
- Fees summary.

3.4 Tests tab

- List of tests: subject, chapter, date, batch, max marks, status (Upcoming, Marks pending, Done), batch average.
- **Create test** button opens a **separate page** (see 3.5).
- **View** button opens a test page: for finished tests it shows batch average, highest, lowest, number below pass mark, ranked results, and the question paper with a Show answers toggle. Pending and upcoming tests show a matching message.
- **Enter marks** and **Edit marks** open a marks sheet.

3.5 Create test page

Fields: subject, batch (or all), date, maximum marks, chapter, and "post on notice board" (ticked by default).

Question paper with AI (optional):

- Teacher enters the chapter, number of questions (up to 20), type (Mixed, MCQ, Short answer) and difficulty, then clicks Generate.
- Each question is saved with its **chapter** and **topic** (a specific sub-topic).
- Teacher can edit questions, answers, marks, chapter and topic, remove questions, add questions manually, and generate more.
- Maximum marks update to the paper total.
- Saving is blocked if any question lacks text, chapter or topic.

3.6 Fees tab

- Tiles: collected this term, pending, number overdue.
- Table: student, batch, paid, balance, due date, status (Paid, Partial, Overdue), Remind, Mark paid.

4. Student screen

- Overall attendance and average.
- One card per subject: latest score, 4-test trend line, change since last test, difference from batch average, subject attendance.
- Chapter and topic analysis (same as 3.3).
- Read-only notice board.

5. Chapter and topic analysis

Hierarchy: Subject > Chapter > Topic (sub-topic). Example: Physics > Thermodynamics > First law.

For each student the system shows:

- **Overall summary by chapter:** good chapters, chapters that need improvement, and the weakest topic to focus on in each subject.
- **Per chapter:** score and level, "Good topics", "Needs improvement" topics, and an expandable table of all topics with score bar, correct out of attempted, and batch average.

Levels: Good is 75% and above. Average is 50% to 74%. Needs improvement is below 50%.

Data needed (the key requirement): results must be recorded per question, or at least per topic, and every question must carry its chapter and topic. Total marks alone cannot tell us where a student is weak. Show "not enough data" until a student has attempted about 5 questions in a topic.

6. Business rules

Rule	Definition
Red flag	A subject dropped 3 tests in a row, or attendance below 80% (thresholds should become configurable)
Trend	Latest overall average vs the previous test: up 2 points or more is Improving, down 2 or more is Dropping
Fee status	Paid: fully paid. Overdue: due date passed with a balance. Partial: balance left, not yet due
Pass mark	40% (assumed)
Attendance	Marked per subject session; the daily figure is derived from it

7. Main data to store

Batch, Student (with parent phone), Teacher, Subject, Chapter, Topic, Session (subject, batch, time), Attendance (per student per session), Test, Question (text, type, options, answer, marks, chapter, topic), Result (per student per question), Fee plan and Payment, Notice, User and role.

Design for many coaching classes on one system (multi-tenant) from the start.

8. AI features

Phase	Feature
Now	AI question paper generation with chapter and topic tags
Next	AI summary of each student for parent reports; plain-language explanation of why a student is flagged
Later	Risk prediction, suggested remedial practice, a chat assistant ("which students need attention this week?")

Rules: a teacher reviews every AI question and answer before use; AI never decides anything about a student on its own. The prototype calls AI through Claude's page runtime. The real product should call an AI service from our own backend, so we control cost, logging and data privacy.

9. Status

In the prototype (sample data, nothing saved): everything in sections 3 to 5 except the items below.

Not built yet:

- Real login and role-based access
 - A real database (reloading resets the demo)
 - Attendance marking screen (the button is a placeholder)
 - Add student form, export report, call and reminder buttons
 - Per-question marks entry
 - Topic scores from real data (they are sample numbers)
 - Print or PDF of question paper
 - WhatsApp or SMS messages
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10. Decisions still open

1. **Marks entry method:** per-question ticks (most accurate, needed for topic analysis) or per-chapter totals (faster, less detail). Suggested: per-question ticks for AI-paper tests, a simple total for other tests.
 2. **Fees:** same for every student, or different by batch, sibling discount or instalments? Should overdue fees also appear on the Overview?
 3. **Notices:** expire automatically after the date? Send by WhatsApp or SMS?
 4. **Red flags:** let the head teacher mark "contacted" with a note? Make thresholds configurable?
 5. **Teacher role:** when to add separate logins for Chemistry and Physics teachers?
 6. **Batches:** are there really several batches, or one? If one, the batch-wise section becomes a single summary.
 7. **Tests:** need start time and duration? Per-batch comparison on the test page?
 8. **Question paper:** print or download as PDF? Should students later take tests on screen, with MCQs auto-marked?
 9. **Students search:** also by phone number or roll number?
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11. Suggested roadmap

Phase	Scope
1. Foundation (about 6 weeks)	Real login and roles, database, add students and batches, attendance marking, test creation, marks entry (per question), notice board
2. Insight	Overview dashboard, red flags, student profile, chapter and topic analysis, fees tab
3. AI and outreach	AI question papers, parent reports, WhatsApp alerts, PDF papers
4. Grow	Teacher role, online tests with auto-marking, risk prediction, second coaching class as a pilot customer

Validate before phase 1 ends: show the prototype to 5 to 10 coaching teachers. Ask what they do today, what annoys them, and whether they would pay.

12. Risks

- **Adoption:** if it is slower than the teacher's current method, they will not use it. Marks entry and attendance must be very fast.
 - **Data quality:** topic analysis is only as good as the tagging and per-question marks. Bad tags give misleading advice.
 - **AI accuracy:** questions and answers can be wrong. Always require teacher review.
 - **Privacy:** we hold data about minors. In India this means following the DPDP Act 2023 (consent, parental consent, security, data deletion).
 - **Trust in flags:** keep every flag explainable, showing the reason.
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13. Suggested tech stack

- **Frontend:** React or Next.js, built as a phone-friendly web app (PWA)
- **Backend:** Node.js or Python (FastAPI)
- **Database:** PostgreSQL
- **Analytics:** plain SQL and Python (pandas) for averages, trends and topic scores
- **AI:** an LLM API, called from the backend only
- **Hosting:** any managed cloud; keep backups and access logs from day one